AutoInsight — Session Report

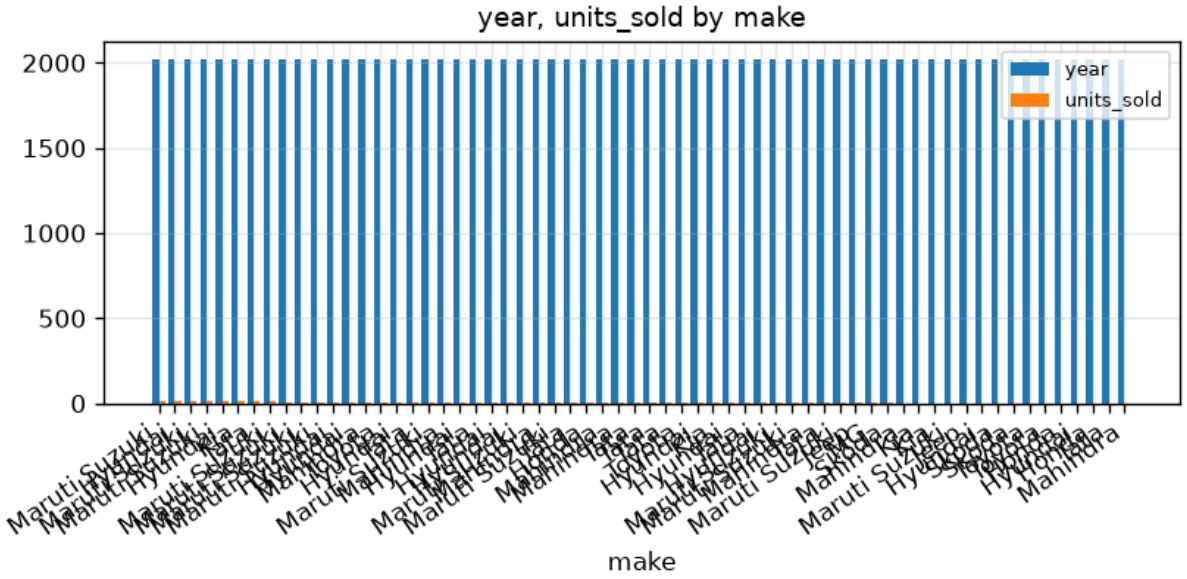
Generated 2026-10-13 19:11:41 · 9 queries in this session

Q1. Which vehicle models sold the most in 2024?

The most sold models in 2024 were Maruti Suzuki Baleno with 16 units, Hyundai i20 with 15 units, and Maruti Suzuki Swift with 11 units.

SQL

```
SELECT v.make, v.model, v.year, COUNT(*) AS units_sold FROM sales AS s JOIN vehicles AS v ON s.vehicle_id = v.id WHERE EXTRACT(YEAR FROM s.sale_date) = 2024 GROUP BY v.make, v.model, v.year ORDER BY units_sold DESC LIMIT 1000
```



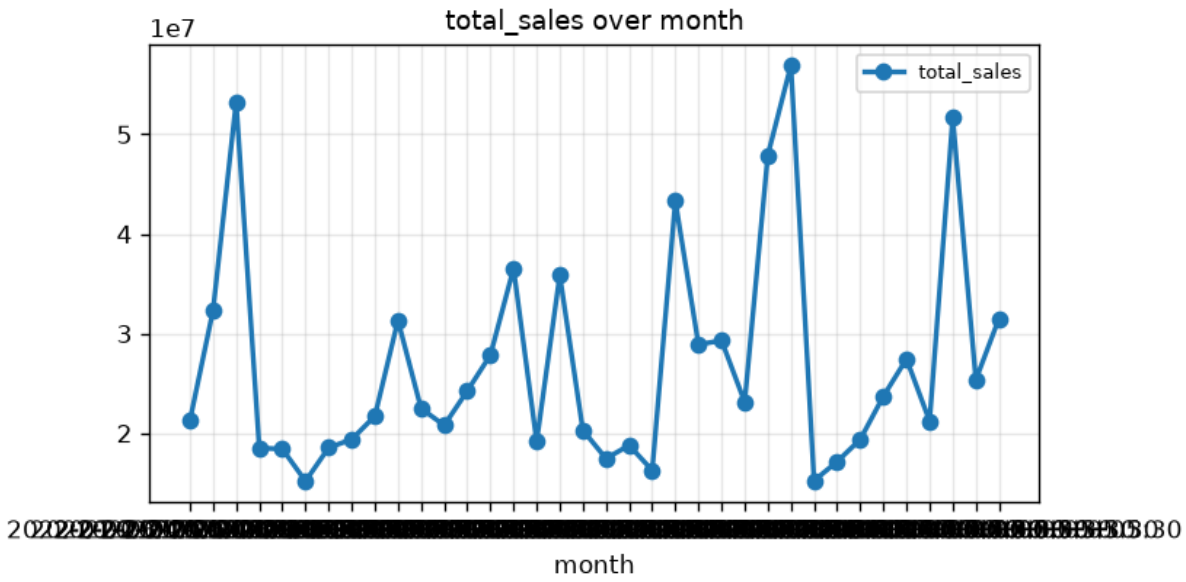
Rows affected: 62

Q2. Show me the monthly sales trend.

The highest monthly sales were in **March 2023** with **■3.65 crore** (■36,53,552.29). The lowest were in **June 2022** with **■1.52 crore** (■15,18,238.28). Overall, sales rose from January 2022 to March 2023, then fluctuated, dipping again in mid■2023.

SQL

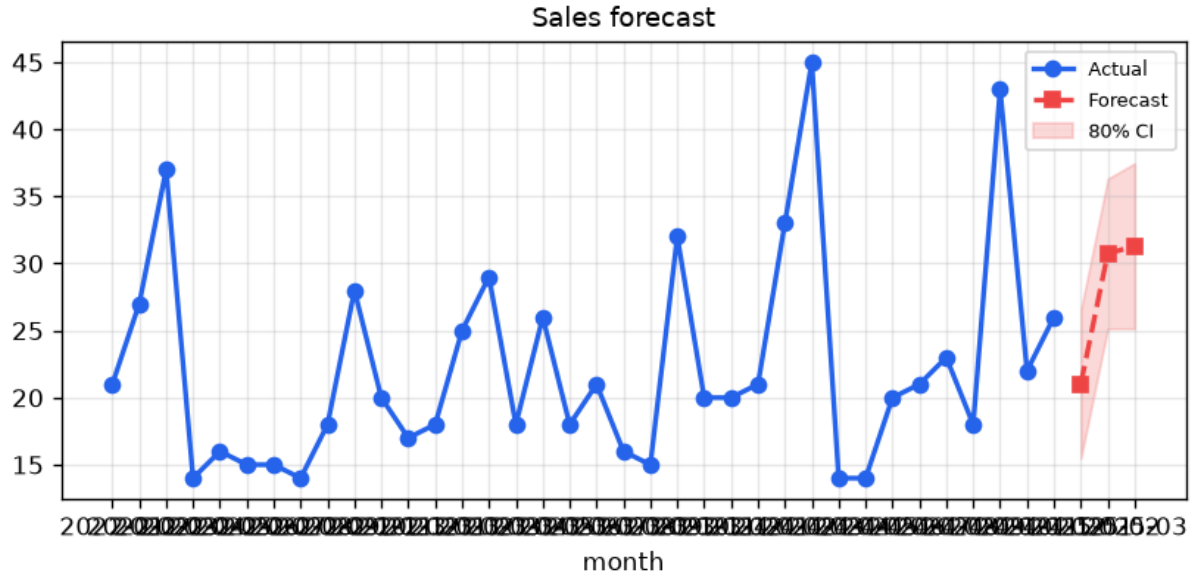
```
SELECT DATE_TRUNC('MONTH', s.sale_date) AS month, SUM(s.sale_price) AS total_sales FROM sales AS s GROUP BY month ORDER BY month LIMIT 1000
```



Rows affected: 36

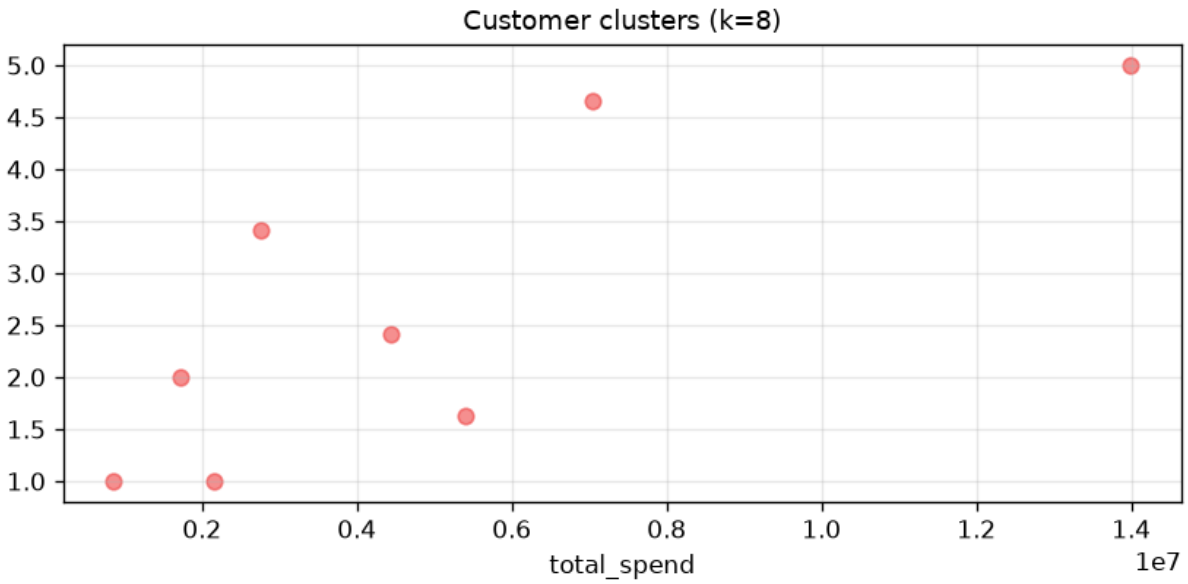
Q3. Forecast sales for the next 3 months.

The forecast shows an overall upward trend, with sales expected to rise from 21 units in January to about 31 units by March. A clear seasonal spike is projected for February, where the predicted sales jump to roughly 31 units, the highest in the three■month window. For the first month ahead (January 2025), the model predicts 21 units, with an 80 % confidence interval ranging from about 15.5 to 26.6 units.



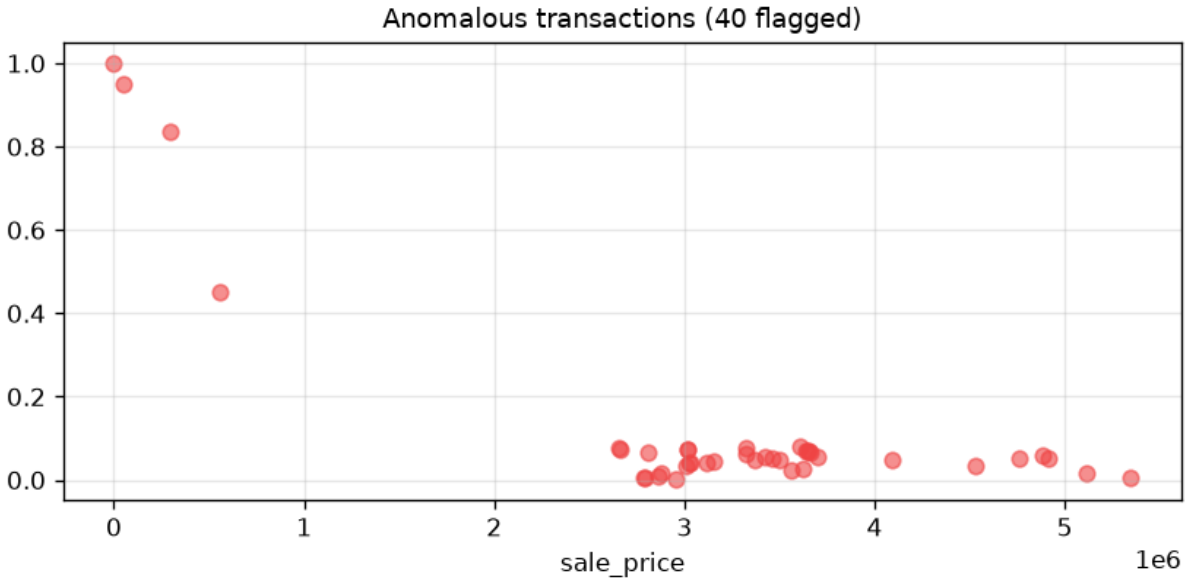
Q4. Segment customers by spending behavior.

The eight■cluster segmentation reveals a clear hierarchy of spending behavior: **Cluster 0** (142 customers) are low■volume, low■spend buyers (■8.6 lakh per purchase), likely first■time or budget shoppers; **Cluster 1** (20 customers) are high■frequency, high■spend repeat buyers (■1.55 lakh average price, ■7.05 crore total spend). **Cluster 2** (62 customers) represent moderate■frequency, moderate■spend buyers (■8.04 lakh average, ■2.75 crore total). **Cluster 3** (36 customers) are single■purchase high■spend customers (■2.15 lakh per purchase, ■2.15 crore total). **Cluster 4** (92 customers) are two■purchase, moderate■spend buyers (■8.60 lakh average, ■1.72 crore total). **Cluster 5** (16 customers) are two■purchase, high■spend buyers (■3.48 lakh average, ■5.40 crore total). **Cluster 6** (36 customers) are two■purchase, high■spend buyers (■1.84 lakh average, ■4.44 crore total). **Cluster 7** (4 customers) are the most valuable repeat buyers, with five purchases averaging ■2.96 lakh and a total spend of ■13.99 crore.


Customer clusters (k=8)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
total_spend
1e7

Q5. Find unusual sales transactions.

The Isolation Forest flagged **40 out of 800** sales transactions as anomalous (5 % contamination). The top offenders feature extreme pricing anomalies: sale 487 shows a ■4,886,058.25 sale price against a ■5,192,410.68 listed price with a 6 % discount; sale 669 records a ■1.00 sale price with a ■1,053,361.99 discount (100 % discount_pct); sale 454 mirrors sale 487's high price■to■discount ratio; sale 297 has a ■48,331.91 sale price with a ■918,306.37 discount (95 % discount_pct); and sale 305 shows a ■300,000.00 sale price with a ■1,537,489.53 discount (84 % discount_pct). These extreme discount percentages and mismatched sale versus listed prices warrant a detailed review by the finance team.

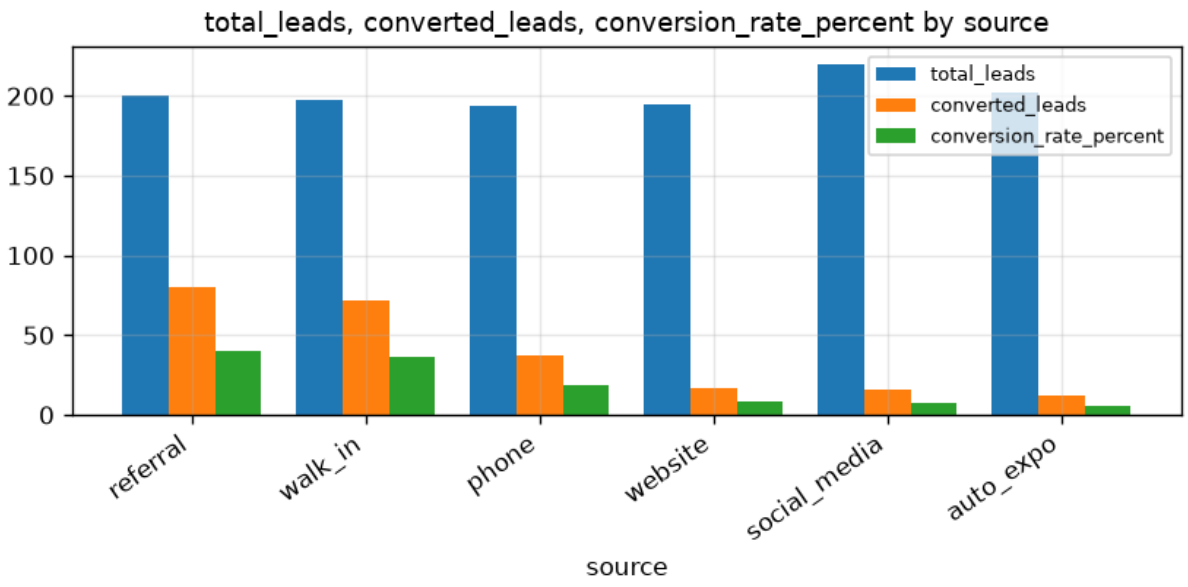


Q6. Analyze lead conversion by source.

The highest lead■to■conversion rate comes from referrals, with 80 out of 200 leads converting – a 40.0 % success rate. Walk■ins follow closely at 36.55 % (72/197), while phone, website, social media and auto■expo sources lag behind, ranging from 19.07 % to 5.94 %. This shows that referral and walk■in channels are the most effective for closing sales.

SQL

```
SELECT l.source, COUNT(*) AS total_leads, SUM(CASE WHEN l.status = 'converted' THEN 1 ELSE 0
END) AS converted_leads, ROUND(CAST(SUM(CASE WHEN l.status = 'converted' THEN 1 ELSE 0 END)
AS DECIMAL) / COUNT(*) * 100, 2) AS conversion_rate_percent FROM leads AS l GROUP BY l.source
ORDER BY conversion_rate_percent DESC LIMIT 1000
```



Rows affected: 6

Q7. How many sales did we make in 2024?

We made **300 sales** in 2024. This figure represents the total count of transactions recorded between January 1 and December 31, 2024.

SQL

```
SELECT COUNT(*) AS total_sales_2024 FROM sales AS s WHERE s.sale_date >= '2024-01-01' AND s.sale_date <= '2024-12-31' LIMIT 1000
```

Rows affected: 1

Q8. List the id, name, city and email of 5 customers from Pune

Here are five customers from Pune: - ID 8: Swati Saxena, swati.saxena@gmail.com - ID 17: Ankita Kaur, ankita.kaur@gmail.com - ID 26: Aarav Sengupta, aarav.sengupta@rediffmail.com - ID 47: Aanya Agarwal, aanya.agarwal@gmail.com - ID 62: Darsh Mukherjee, darsh.mukherjee@rediffmail.com

SQL

```
SELECT id, name, city, email FROM customers WHERE city = 'Pune' LIMIT 5
```

Rows affected: 5

Q9. List all sales made in the year 2031

There were no sales recorded in the year 2031.

SQL

```
SELECT * FROM sales AS s WHERE EXTRACT(YEAR FROM s.sale_date) = 2031 LIMIT 1000
```

Rows affected: 0